TimeMultiformer: Attention-Based Collaborative Feature and Temporal Dependencies Learning for Multivariate Time Series Imputation

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Abstract. Multivariate time series data generated in fields such as transportation, economics, and healthcare can support various analytical tasks. However, due to the common occurrence of missing data in real-world scenarios, existing work faces significant challenges in ensuring data interpretability and effectively conducting specific tasks like advanced analysis and pattern recognition. Traditional missing value imputation methods struggle to effectively capture temporal dependencies in time series data. Although deep learning-based approaches (e.g., Transformer) demonstrate advantages in modeling long-range dependencies, they generally suffer from two critical limitations: (1) inadequate consideration of unequal time intervals between adjacent observations; (2) deficiencies in joint modeling of temporal and feature dependencies. To address these gaps, this paper proposes an attention-based framework named TimeMultiformer. The framework innovatively integrates Transformer and iTransformer architectures to accurately capture temporal dependencies and inter-variable feature dependencies, respectively. Specifically, it introduces a time lag matrix to explicitly model the impact of historical observations on current imputation and employs a diagonal-masked multi-head attention mechanism to mitigate the model's over-reliance on current time-step information, thereby enhancing the capture of long-range dependencies. Additionally, through a joint data imputation learning strategy, TimeMultiformer exhibits superior performance in time series imputation tasks. Extensive experiments on four real-world datasets demonstrate that TimeMultiformer outperforms state-of-the-art methods in time series imputation both quantitatively and qualitatively, with the mean absolute error (MAE) reduced by 2%–16%, root mean square error (RMSE) reduced by 10%–26%, and the average reduction of mean relative error (MRE) exceeding 10%. The source code of this project is publicly available at: https://github.com/wzq-come-on/TimeMultiformer.git.

Keywords: multivariate time series · Transformer · iTransformer · Missing data imputation

1 INTRODUCTION

Multivariate time series [28] are ubiquitous in the real world, being extensively generated and utilized across diverse domains including transportation science [10] and meteorology [18, 24]. Each time series not only possesses temporal characteristics but also contains variables that may be interrelated or independent. Through the analysis and mining of multivariate time series, associations and patterns between different variables can be revealed, providing more comprehensive information for decision-making. However, missing values (MVs) are introduced into multivariate time series due to sensor failures during data acquisition, communication errors, and unexpected malfunctions. These MVs severely compromise data interpretability and pose challenges to advanced analytical tasks such as classification and clustering. According to statistics, over 40% of datasets provided by the commonly used UCI [1] database are incomplete.

With the rapid advancements in deep learning and neural networks in recent years, numerous deep learning-based time series imputation methods have been proposed, such as GAIN [25], E^2GAN [14], DeepMVI [2], BRITS [3], etc. Notably, in the field of time series data processing, Transformer can directly model dependency relationships between any positions in a sequence (regardless of their temporal distance) through its self-attention mechanism. This characteristic grants the Transformer unique advantages in handling time series data. Moreover, Transformer effectively leverages both global information (e.g., historical trends) and local context (e.g., adjacent data points) for imputation inference. This dual capability enables the Transformer to excel in time series imputation, offering more accurate and reliable results.

Owing to its superior capability in capturing global dependencies, various Transformer-based time series imputation models have been proposed, such as Informer [29] and SAITS [7]. However, most existing approaches face two limitations: (1) **It does not account for the issue of unequal time intervals between two valid adjacent observations.** In time series data, observations are not always recorded at consistent intervals, leading to variable time lags between two complete observations. This variability implies that the influence of past observations on current values should be adjusted based on the length of these time lags. Therefore, it is essential to consider these changing time lags to accurately model the temporal dynamics and ensure that the influence of past observations is appropriately weighted according to the actual time elapsed. (2) **It inadequately explores the feature dependencies within the multivariate time series.** While most existing works have extensively developed methods to capture temporal dependencies, the importance of feature dependencies has often been overlooked. However, the feature dependencies are crucial for accurate multivariate imputation, as they can significantly enhance the understanding of complex relationships between different variables and improve the overall quality of the imputation process.

[1] https://archive.ics.uci.edu/

To address these issues, we propose a novel TimeMultiformer framework for multivariate time series imputation. The proposed TimeMultiformer integrates Transformer and iTransformer architectures, where the Transformer component captures temporal dependencies while the iTransformer models the inter-variable (feature) dependencies. Moreover, we introduce a time lag matrix, as detailed in Definition 4, to serve as an input for the TimeMultiformer model. This matrix allows the model to apply a variable influence from past data, effectively accommodating the changing time intervals between observations. We validated the effectiveness of our model on four real-world datasets, demonstrating that it achieves substantial improvements in both imputation tasks and downstream classification tasks compared to previous models. Our main contributions are summarized as follows:

- We propose TimeMultiformer, a novel multivariate time series imputation model that comprehensively leverages attention mechanisms.The proposed model, built upon Transformer and iTranformer, can more accurately capture both temporal and feature dependencies.
- We model a time lag matrix to reflect the impact of the past observed values on the current imputation. By including the time lag matrix as part of TimeMultiformer's input, the model can effectively capture changing temporal correlations among observations.
- We conducted extensive experiments on four real-world datasets, quantitatively and qualitatively evaluating TimeMultiformer to validate the effectiveness of the proposed algorithm.

2 RELATE WORK

In recent years, various missing value imputation methods have been proposed. However, due to space constraints, this paper focuses solely on the most directly relevant works for time series imputation. Based on the typical models, we classify the existing works into the following categories: (1) RNN-based imputation methods, e.g., GRU-D [5], BRITS [3], and M-RNN [26], excel at capturing temporal information for time series imputaiton. But they are difficult to model long-term temporal dependencies, and thereby degrade performance. (2) GAN-based imputation methods, e.g., GAIN [25], E^2GAN [14], and NAOMI [12], explore the generability of the GANs for time series imputation. However, GANs commonly encounter non-convergence and mode collapse issues, necessitating the design of refined training pipelines to stabilize the adversarial learning process. (3) VAE-based imputation methods, e.g., MIVAE [16], GP-VAE [8], and MIWAE [17], enhance the capture of complex spatiotemporal dependencies through probabilistic modeling. While they have inherent advantages in explicit probabilistic modeling, their generative capability is constrained by the representational power of the latent space. (4) Attention-based imputation methods, e.g., CDSA [15], DeepMVI [2], and SAITS [7], utilize self-attention techniques to model time series. Attention-based models usually outperform other models due

to their advantages in handling remote dependencies and parallel processing capabilities. However, they increase the computational cost when dealing with long sequences, which can lead to slower training and inference.

3 PROBLEM STATEMENT

In this section, we first introduce the notations and definitions used in this paper and describe our research objectives.

Definition 1. *We denote the multivariate time series as* $\mathbf{X} = (\mathbf{x}_{t_1}, \mathbf{x}_{t_2}, \cdots, \mathbf{x}_{t_n}) \in \mathbb{R}^{n \times d}$*, representing a sequence of observations arriving at time points* $\mathbf{T} = (t_1, t_2, \cdots, t_n)$*. Each* $\mathbf{x}_{t_i}$ *is a d-dimensional observation of* $\mathbf{X}$*, and* $x^j_{t_i}$ *denotes the* j*th variable of* $\mathbf{x}_{t_i}$*. Notably, if any observation* $x^j_{t_i}$ *is missing, the time series* $\mathbf{X}$ *is considered incomplete.*

Definition 2. *The mask matrix* $\mathbf{M} \in \mathbb{R}^{n \times d}$ *is defined to indicate the missingness of the multivariate time series* $\mathbf{X}$*. Each entry* $m^j_{t_i}$ *in* $\mathbf{M}$ *corresponds to the presence or absence of the value* $x^j_{t_i}$*, defined such that* $m^j_{t_i} = 1$ *when* $x^j_{t_i}$ *is observed and* $m^j_{t_i} = 0$ *otherwise.*

Definition 3. *The artificial mask matrix* $\mathbf{I} \in \mathbb{R}^{n \times d}$ *is used to characterize the status of artificial masking for missing values in the multivariate time series* $\mathbf{X}$*. Each element* $i^j_{t_i}$ *in the matrix* $\mathbf{I}$ *corresponds to indicating whether the value* $x^j_{t_i}$ *is artificially masked, defined such that* $i^j_{t_i} = 1$ *when* $x^j_{t_i}$ *is artificially masked, and* $i^j_{t_i} = 0$ *otherwise.*

In this paper, we assume that the data missing mechanism follows Missing Completely at Random (MCAR), meaning that any feature dimension $x^j_{t_i}$ (where $1 \leq j \leq d$) at time point $t_i \in \mathbf{T}$ may exhibit missingness. Notably, due to the presence of Missing Values (MV), the time intervals between consecutive valid observations exhibit dynamic variability. Such variable time delay information serves as a critical indicator for characterizing the strength of temporal correlations. Thus, we define a time lag matrix followed by [14].

Definition 4. *A time lag matrix* $\Delta \in \mathbb{R}^{n \times d}$ *indicates the time intervals between two adjacent observations, where the mathematical formulation for each individual time lag* $\delta^j_{t_i}$ *within* $\boldsymbol{\Delta}$ *is defined as:*

$$\delta^j_{t_i} = \begin{cases} t_i - t_{i-1} & \text{if } m^j_{t_{i-1}} = 1 \,\&\, i > 0 \\ \delta^j_{t_{i-1}} + t_i - t_{i-1} & \text{if } m^j_{t_{i-1}} = 0 \,\&\, i > 0 \\ 0 & \text{if } i == 0 \end{cases} \tag{1}$$

MTS Imputation. Given an incomplete multivariate time series $\mathbf{X}$, the MTS imputation is to generate a complete time series $\widetilde{\mathbf{X}}$ by replacing the missing values in $\mathbf{X}$ with plausible ones (close to the ground truth) through an MV imputation model $\mathbf{M}(\mathbf{X}, \mathbf{M}, \boldsymbol{\Delta})$.

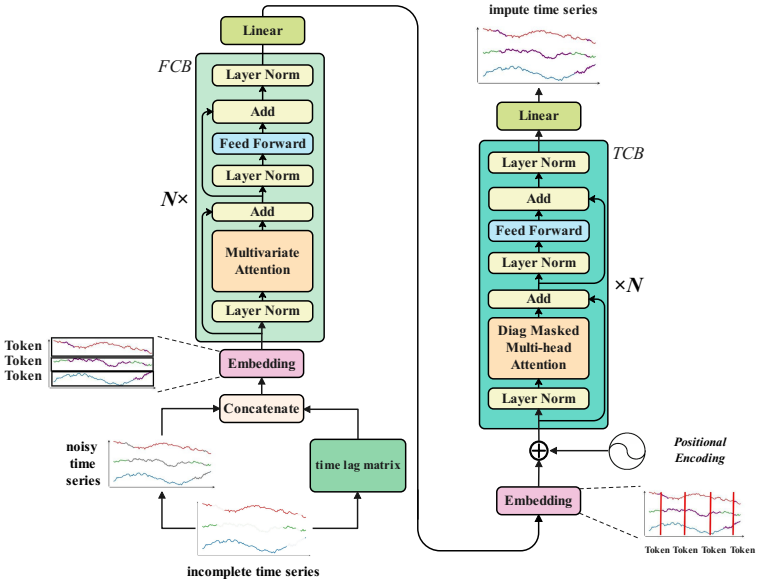

Fig. 1. Architectural Overview of the TimeMultiformer ModelThe framework consists of two core components: the Feature Dependencies Learning Net (FDLN) and the Temporal Dependencies Learning Net (TDLN).

4 METHODOLOGY

In this section, we first outline the proposed TimeMultiformer. Then, we present the details of each component of the model.

4.1 TimeMultiformer-an Overview

As shown in Fig 1, the proposed TimeMultiformer employs a collaborative architecture of Transformer and iTransformer to construct a Feature Dependencies Learning Net (FDLN) and a Temporal Dependencies Learning Net (TDLN). It overcomes the limitations of traditional time-series modeling methods through a dual-pathway modeling mechanism. TimeMultiformer first utilizes FDLN to learn feature dependencies. Taking the time series of each feature as input, this module addresses the issue of data misalignment caused by unequal sampling intervals of adjacent observations by introducing a time lag matrix as an additional input. Subsequently, TDLN is employed, with the observations composed of all features at each timestamp serving as input, to learn temporal dependencies. Moreover, to mitigate the model's over-reliance on the current time-step information during data imputation, we modify the Transformer by adopting a diagonal masking mechanism [19]. By suppressing the diagonal elements of the attention map, the model is compelled to focus on other time steps, thereby

enabling precise capture of temporal dependencies and feature correlations and enhancing imputation performance.

Compared with the standard Transformer [22], which incurs information loss in the feature dimension due to global attention calculations along the sequence dimension, and the iTransformer [11], which struggles to capture long-range temporal patterns when performing attention mapping in the feature dimension, this architecture innovatively enables the modeling of cross-variable correlations and temporal dependencies.

4.2 Feature Dependencies Learning Net (FDLN)

Network Input: Given an incomplete multivariate time series $\mathbf{X}$, the FDLN takes as input the noise matrix $\widetilde{\mathbf{X}}$ and the time lag matrix Δ computed from $\mathbf{X}$. Specifically, we first generate a random noise matrix $\mathbf{Z} \in \mathbb{R}^{n \times d}$, where each element $z_{t_i}^j \sim \mathcal{N}(0,1)$, and fill the missing values in $\mathbf{X}$ with entries from $\mathbf{Z}$, such that $\widetilde{\mathbf{X}} = \mathbf{M} \odot \mathbf{X} + (1 - \mathbf{M}) \odot \mathbf{Z}$ (unlike most approaches that directly fill missing values with zeros). This noise matrix aids the model in learning more effectively.

Network Architecture: As illustrated in Fig 1, the FDLN comprises an Embedding layer and N stacked Feature Capture Blocks (FCBs), where each block consists of self-attention, feed-forward network, normalization layer, etc. The architecture closely resembles that of the iTransformer encoder. Then, we concatenate $\widetilde{\mathbf{X}}$ and Δ, feed them into the embedding layer, and obtain the FDLN input $\mathbf{X}_I$ as shown below.

$$\mathbf{X}_I = \text{Embedding}(\text{Concat}(\overline{\mathbf{X}}, \Delta)) \tag{2}$$

Secondly, as previously described, the network contains N stacked FCBs. Inspired by Vaswani et al., the FCB innovatively treats the full time series of each variable as independent processes. We take $\mathbf{X}_I$ as the input to the FCBs and obtain $\mathbf{X}_F$, as shown in the following equation:

$$\mathbf{X}_F = \text{FCBs}(\mathbf{X}_I) \tag{3}$$

Notably, the task of generating the predicted sequence is essentially delegated to linear layers — which has been proven effective by previous work [6]. Therefore, we ultimately utilize linear layers to generate the imputation results of the Feature Dependencies Learning Network (FDLN). The formula is as follows:

$$\mathbf{X}_{FDLN} = LN(\mathbf{X}_F) \tag{4}$$

4.3 Temporal Dependencies Learning Net (TDLN)

Network Input: In this study, the imputed matrix generated by the Feature Dependencies Learning Net $\mathbf{X}_{FDLN}$ serves as the input to the Temporal Dependencies Learning Net.

Network Architecture: As shown in the Fig 1, the Temporal Dependencies Learning Net comprises a Positional encoding layer, N Temporal Capture Blocks(TCBs), and a linear layer, with an architecture closely resembling that of the Transformer.

Positional encoding: Since the self-attention mechanism lacks inherent awareness of the position or order of each observation, a positional encoding layer is introduced to provide the model with information about the positions of observations in the time series. Incorporating relative positional encoding facilitates capturing temporal dependencies and feature correlations between observations.

Specifically, for the i-th observation x_{ti} in the time series X, the computation of its positional encoding is as follows:

$$\mathrm{PE}(\mathbf{x}_{t_i}) = \begin{cases} sin(w_k \cdot i), & \text{if} i == 2k \\ cos(w_k \cdot i), & \text{if} i == 2k+1 \end{cases} \tag{5}$$

where $w_k = \frac{1}{10000^{\frac{2k}{d_m}}}$, and d_m represents the number of dimensions of the exported position encoding.

Temporal Capture Blocks: As described previously, the Temporal Dependencies Learning Net contains N Temporal Capture Blocks. Each block is composed of a diagonally masked multi-head attention sub-block, a feed-forward sub-block, as well as components like the layer normalization layer and the residual connection layer. Inspired by SAITS [7], we employ diagonally masked multi-head attention (DiagMaskedMHA)—an improvement based on the single-head attention mechanism proposed by Vaswani et al. [22] in 2017—to replace traditional multi-head attention. The specific formula for DiagMaskedMHA is as follows:

$$\begin{aligned} &\mathrm{DiagMaskedMHA}(Q, K, V) \\ &\qquad = Concat(head_1, head_2, \cdots, head_h)W^O \end{aligned} \tag{6}$$

where Q is the query vector, K is the key vector, and V is the value vector, which respectively correspond to dimensions d_k, d_k, and d_v. Additionally, h denotes the number of heads, and W^O is the output transformation matrix. The output of the i-th head, head_i, is defined as follows:

$$\begin{aligned} head_i &= \mathrm{DiagMaskedAttn}(Q_i, K_i, V_i) \\ &= \mathrm{Softmax}\left(\mathrm{DiagMask}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right)\right) V_i \end{aligned} \tag{7}$$

where $Q_i = QW_i^Q$, $K_i = KW_i^K$, and $V_i = VW_i^V$, with (W_i^Q, W_i^K, W_i^V) denoting the weight matrices for the i-th head's (Q, K, V), respectively. Unlike traditional MHA, where $\mathrm{Head}_i = \mathrm{Softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V$, DiagMaskedMHA modifies the design by setting the diagonal values of the score matrix $\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right)$ to $-\infty$, ensuring that

the corresponding weights approach zero after applying the softmax function. The specific functional formula for DiagMaskedMHA is as follows:

$$\text{DiagMask}\left(\frac{QK^T}{\sqrt{d_k}}\right) = \left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot A_m \tag{8}$$

$$A_m(i,j) = \begin{cases} -\infty & \text{if} i = j \\ 1 & \text{if} i \neq j \end{cases} \tag{9}$$

The DiagMask operation ensures that the estimation of missing values (MV) depends solely on other observations, thereby preventing them from being influenced by their own estimation.

The DiagMaskedMHA module proposed in this study achieves the joint capture of temporal dependencies and feature correlations through a single attention operation. However, the internal structure of the classical multi-head attention mechanism, dominated by linear transformations, exhibits limitations in representational capacity when modeling complex data patterns compared to nonlinear transformations.

To address this, we introduce a feed-forward sub-block after the DiagMaskedMHA sub-module, enhancing the overall representational learning efficiency of the model through nonlinear enhancement.

The Feed-Forward sub-block designed in this study adopts a two-layer cascaded fully connected architecture, characterized by a position-aware independent feature evolution mechanism. Specifically, the module consists of two fully connected layers with independent parameters, where each neuron responds exclusively to the input vector at the corresponding temporal position—i.e., performing parallel and independent processing (position-wise processing) on the feature vector of each time step in the input sequence. This design preserves the integrity of local temporal features while enabling deep reconstruction of the feature space through nonlinear transformations. Given the input feature vector $\mathbf{x}_f \in \mathbb{R}^d$ to the current feed-forward sub-module, its specific definition is as follows:

$$\text{FFN}(\mathbf{x}_f) = \text{ReLU}(\mathbf{x}_f \cdot \mathbf{W}_1 + \mathbf{b}_1) \cdot \mathbf{W}_2 + \mathbf{b}_2 \tag{10}$$

where W_1, W_2, b_1, and b_2 are learnable parameters. As shown in the formula, the feed-forward sub-block plays a crucial role in learning the output distribution, thereby facilitating time series imputation.

Additionally, layer normalization and residual connections are key components of the Transformer. The formula for layer normalization is as follows:

$$\text{LayerNorm}(\mathbf{x}_L) = \frac{\mathbf{x}_L - \mu}{\sigma} \cdot \mathbf{W}_L + \mathbf{b}_L \tag{11}$$

where μ and σ denote the mean and standard deviation of $\mathbf{x}_L$, respectively, and W_L and b_L are trainable parameters.

Residual connections serve primarily to optimize model training by duplicating the information from the previous layer: one copy is directly transmitted to the next layer, while the other is processed through the current layer. The two

results—i.e., the direct transmission and the transformed output of the current layer—are summed to form the input for the next layer. This design ensures that at least the information from the previous layer is preserved, thereby enhancing model performance.

The data processing flow of TDLN can be summarized as follows:

$$\mathbf{X}_{PE} = \mathrm{PE}(\mathbf{X}_{FDLN}) \tag{12}$$

$$\mathbf{X}_{S} = \mathrm{TCBs}(\mathbf{X}_{PE}) \tag{13}$$

$$\widehat{\mathbf{X}} = \mathbf{X}_{TDLN} = LN(\mathbf{X}_{S}) \tag{14}$$

Among them, in Eq 12, $\mathbf{X}_{PE}$ represents the output of $\mathbf{X}_{FDLN}$ after positional encoding. $\mathbf{X}_{S}$ in Eq 13 denotes the output after passing through N Temporal Capture Blocks. $\widehat{\mathbf{X}}$ in Eq 14 signifies the final output of the entire model.

4.4 Loss Functions of Learning Objectives:

To facilitate model training, we propose a joint data imputation learning task based on SAITS [7]. As illustrated in the figure, the joint data imputation learning task consists of an Observed Value Imputation Task and an Artificial Mask Imputation Task. Both tasks employ the MAE loss function defined in Eq.19 (denoted as ℓ_{MAE}), which takes three input parameters: estimation, target, and mask and calculates the mean absolute error (MAE) between the estimation and target at positions indicated by the mask.

As shown in Eq.15, $\mathcal{L}_{\mathrm{OVIT}}$ is calculated by summing the losses of $\mathbf{X}_{FDLN}$ and $\mathbf{X}_{TDLN}$. To prevent it from dominating the gradient direction due to excessively large magnitude, its final value is normalized by a weighting factor of 2 (i.e., taking the arithmetic mean of the two). As shown in Eq.16, $\mathcal{L}_{\mathrm{AMIT}}$ is the loss calculated over the mask vector I for the final imputed matrix $\widehat{\mathbf{X}}$. As shown in Equation 17, the final loss $\mathcal{L}$ is obtained by summing $\mathcal{L}_{\mathrm{OVIT}}$ and $\mathcal{L}_{\mathrm{AMIT}}$ in a certain proportion. Here, μ and λ serve as hyperparameters that regulate the weights of each component of the loss function.

$$\mathcal{L}_{\mathrm{OVIT}} = \frac{1}{2}\left(\ell_{\mathrm{MAE}}\left(\mathbf{X}_{FDLN}, \mathbf{X}, \hat{M}\right) + \ell_{\mathrm{MAE}}\left(\mathbf{X}_{TDLN}, \mathbf{X}, \hat{M}\right)\right) \tag{15}$$

$$\mathcal{L}_{\mathrm{AMIT}} = \ell_{\mathrm{MAE}}\left(\widehat{\mathbf{X}}, \mathbf{X}, I\right) \tag{16}$$

$$\mathcal{L} = \mu\mathcal{L}_{\mathrm{OVIT}} + \lambda\mathcal{L}_{\mathrm{AMIT}} \tag{17}$$

4.5 Imputation:

Given an incomplete time series $\mathbf{X}$, we reconstruct the complete time series $\hat{\mathbf{X}}$ using TimeMultiformer. With the joint data imputation learning mechanism, the estimates at corresponding positions in $\hat{\mathbf{X}}$ are ultimately used to impute the missing parts in $\mathbf{X}$, and the resulting complete imputed time series is denoted as:

$$\tilde{X} = \mathbf{X} \odot \mathbf{M} + \widehat{\mathbf{X}} \odot (1 - \mathbf{M}) \tag{18}$$

5 EXPERIMENTS

5.1 Experimental Settings

Datasets: We adopt four real-world time series datasets, i.e., PhysioNet-2012 (PhysioNet 2012 Mortality Prediction Challenge) [9], Air-Quality (Beijing Multi-Site Air-Quality) [27], Electricity (Electricity Load Diagrams) [21], and ETT [1] (Electricity Transformer Temperature) to conduct experiments.

Algorithms for comparison: We evaluate the proposed TimeMultiformer against nine state-of-the-art time series imputation algorithms, including Median, Last, GRUI-GAN [13], E^2GAN [14], M-RNN [26], GP-VAE [8], BRITS [3], Transformer [22], SAITS [7] and MTSCI [30]. The Median and Last are two naive imputation methods in which missing values (MVs) are replaced by the median of the training set and the last observed value, respectively. GRUI-GAN and E^2GAN are state-of-the-art GAN-based time series imputation methods proposed by Luo et al. [13, 14].M-RNN, GP-VAE, and BRITS are autoregressive imputation models that leverage RNNs to capture temporal correlations. Transformer and SAITS represent the latest attention-based time series imputation frameworks, which process input data in parallel and globally.

Model Settings: We implement the proposed TimeMultiformer using PyTorch, building extensively upon established methodologies for model architecture design. The hyperparameter settings of TimeMultiformer are detailed in Table 1. Here, the parameter *n_groups* denotes the number of Temporal Capture Blocks, depth represents the number of Feature Capture Blocks, and *n_head* stands for the number of heads in the multi-head attention mechanism. Although the model achieves satisfactory performance with the final parameter configuration described in this table, there remains potential for further optimization. All models were trained with the Adam optimizer on NVIDIA RTX A6000 GPUs.

Measurements: We employ three evaluation metrics—mean absolute error (MAE), root mean square error (RMSE), and mean relative error (MRE)—to assess the missing value (MV) imputation performance of different models. Notably, imputation errors are calculated exclusively for the missing values specified by the mask matrix $\mathbf{M}$.

$$\mathrm{MAE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right) \cdot m_{t_i}^j \right|}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} m_{t_i}^j} \tag{19}$$

$$\mathrm{RMSE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \sqrt{\frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left[\left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right)^2 \cdot m_{t_i}^j \right]}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} m_{t_i}^j}} \tag{20}$$

$$\mathrm{MRE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right) \cdot m_{t_i}^j \right|}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \widetilde{x}_{t_i}^j \cdot m_{t_i}^j \right|} \tag{21}$$

Table 1. Hyper-parameters Settings

Hyper-parameters	PhysioNet-2012	Air-Quality	ETT(m1)	Electricity
batch_size	128	128	128	128
lr	0.0005	0.0001	0.001	0.0002
d_k	32	128	32	128
d_model	512	1024	1024	1024
d_v	64	64	32	128
epochs	1000	10000	10000	1000
n_group	2	2	2	5
depth	1	1	2	2
n_head	8	4	1	8

5.2 Comparison with Baselines

First, we present the imputation errors measured by three metrics (MAE, RMSE, and MRE) on four datasets with a 10% missing rate. As shown in Table 2, recent deep-learning-based imputation models generally outperform the Median and Last methods, which validates the effectiveness of deep learning models in time series imputation. Additionally, in most scenarios, attention-based models (e.g., Transformer and SAITS) generally surpass pure generative adversarial network (GAN)-based models (i.e., GRUI-GAN and E^2GAN) and recurrent neural network (RNN)-based models (i.e., M-RNN, GP-VAE, and BRITS). The TimeMultiformer model proposed in this paper, constructed upon Transformer and iTransformer, demonstrates robust performance by accurately capturing temporal and feature dependencies, achieving a general performance improvement of approximately 10% compared to other attention-based models. Due to the high dimensionality and long time series characteristics of the Electricity dataset, GRUI-GAN and E^2GAN encounter the problem of loss explosion and thus fail to produce interpolation results. When running the MTSCI based on diffusion models, it brings an extremely high computational load, leading to excessive GPU resource occupation and long operation time. Therefore, the relevant experimental results are not presented on this dataset for the time being.

Next, to further demonstrate the superiority of TimeMultiformer, Table 3 presents the imputation performance of several state-of-the-art models on the Electricity dataset across different missing rates (20%∼70%). It is obvious that the imputation errors of all deep learning-based imputation models gradually increase as the missing rate rises, which is because there is less valid information available for multivariate imputation.

5.3 Applications in Classification

Given the widespread presence of missing values (MVs) in real-world datasets, tasks such as classification and pattern recognition face significant challenges. To

Table 2. Performance comparison for Imputation

Method	PhysioNet-2012	Air-Quality
Median	0.725 / 0.985 / 103.1%	0.761 / 1.173 / 107.2%
Last	0.850 / 1.188 / 118.3%	0.948 / 1.389 / 129.1%
GRUI-GAN	0.760 / 1.021 / 107.0%	0.766 / 1.155 / 108.1%
E^2GAN	0.698 / 0.944 / 99.1%	0.752 / 1.124 / 105.7%
M-RNN	0.521 / 0.765 / 74.9%	0.302 / 0.651 / 42.0%
GP-VAE	0.410 / 0.655 / 57.1%	0.273 / 0.632 / 38.3%
BRITS	0.259 / 0.780 / 36.5%	0.159 / 0.554 / 22.6%
Transformer	0.197 / 0.466 / 27.5%	0.168 / 0.571 / 23.3%
SAITS	0.186 / 0.431 / 26.6%	0.143 / 0.336 / 20.3%
MTSCI	**0.142** / 0.389 / **24.6%**	0.140 / 0.408 / 19.7%
TimeMultiformer	0.184 / **0.319**/ 27.8%	**0.118 / 0.303 / 16.7%**
Method	**ETT(m1)**	**Electricity**
Median	1.151 / 1.862 / 139.9%	2.088 / 2.739 / 112.1%
Last	0.977 / 1.257 / 95.9%	0.988 / 1.497 / 51.3%
GRUI-GAN	0.603 / 0.710 / 94.2%	/
E^2GAN	0.558 / 0.688 / 87.0%	/
M-RNN	0.389 / 0.446 / 32.1%	1.231 / 1.848 / 65.8%
GP-VAE	0.277 / 0.322 / 15.9%	1.113 / 1.591 / 58.9%
BRITS	0.133 / 0.264 / 12.8%	0.881 / 1.333 / 46.3%
Transformer	0.121 / 0.193 / 11.8%	0.877 / 1.327 / 45.6%
SAITS	0.098 / 0.151 / 9.3%	0.796 / 1.232 / 42.6%
MTSCI	0.238 / 0.379 / 18.6%	/
TimeMultiformer	**0.084 / 0.125 / 8.0%**	**0.572 / 0.945 / 30.6%**

further evaluate the effectiveness of TimeMultiformer, we conducted a real-world data classification experiment using the PhysioNet-2012 dataset. The rationale of the experimental design is that improved input data quality (specifically, the imputation results generated by our model) should enhance the performance of subsequent classification tasks, with the imputed data serving as the input for these tasks. We developed an RNN-based classifier with configurations identical to those used in SAITS.

Fig. 2 reports the classification results of data imputed by existing methods and data processed by TimeMultiformer (we use three metrics to evaluate the classification results: ROC-AUC [4], PR-AUC [20], and F1-score [23]). The results show that TimeMultiformer achieves the best performance across all metrics. Compared with the latest state-of-the-art (SOTA) model, it significantly improves ROC-AUC, PR-AUC, and F1-score by 0.7%, 3.8%, and 1.4%, respec-

Table 3. Performance comparison of different methods on the Electricity dataset across varying missing rates from 20% to 70% is presented. The reported metrics are MAE / RMSE / MRE, where lower values indicate better performance.

Method	20%	30%	40%
GP-VAE	1.120 / 1.497 / 59.9%	1.012 / 1.540 / 56.2%	1.090 / 1.567 / 58.1%
BRITS	0.933 / 1.412 / 49.9%	0.942 / 1.433 / 50.4%	0.988 / 1.489 / 53.9%
Transformer	0.851 / 1.325 / 45.4%	0.857 / 1.331 / 45.6%	0.878 / 1.389 / 46.9%
SAITS	0.763 / 1.187 / 40.8%	0.790 / 1.223 / 42.3%	0.869 / 1.314 / 46.7%
TimeMultiformer	**0.587 / 0.999 / 31.4%**	**0.725 / 1.106 / 40.2%**	**0.833 / 1.274 / 45.1%**
Method	**50%**	**60%**	**70%**
GP-VAE	1.089 / 1.567 / 58.6%	1.103 / 1.621 / 59.1%	1.042 / 1.601 / 55.7%
BRITS	1.021 / 1.528 / 55.1%	1.103 / 1.606 / 59.1%	1.092 / 1.611 / 58.6%
Transformer	0.901 / 1.418 / 48.2%	0.898 / 1.412 / 48.1%	0.908 / 1.432 / 48.7%
SAITS	0.876 / 1.377 / 46.9%	**0.892** / 1.328 / **47.9%**	**0.898** / 1.273 / **48.1%**
TimeMultiformer	**0.875 / 1.312 / 46.9%**	0.903 / **1.283** / 48.5%	0.907 / **1.263** / 48.8%

Table 4. Ablation Experiment results with each part of modifications

Model	PhysioNet-2012	Air-Quality
Base	0.304 / 0.466 / 45.8%	0.129 / 0.330 / 18.3%
Base + DiagMask	0.298 / 0.447 / 45.0%	0.129 / 0.330 / 18.3%
Base + DiagMask + Δ	0.319 / 0.474 / 48.1%	0.128 / 0.316 / 18.2%
TimeMultiformer	**0.189 / 0.327 / 28.5%**	**0.118 / 0.303 / 16.7%**
Model	**ETT(m1)**	**Electricity**
Base	0.129 / 0.200 / 12.3%	0.864 / 1.379 / 46.3%
Base + DiagMask	0.159 / 0.244 / 15.2%	0.845 / 1.331 / 45.3%
Base + DiagMask + Δ	0.127 / 0.196 / 12.1%	0.848 / 1.2583 / 45.5%
TimeMultiformer	**0.084 / 0.130 / 8.0%**	**0.572 / 0.945 / 30.6%**

tively. The results demonstrate the superiority and general applicability of the proposed TimeMultiformer.

5.4 Ablation Study

To improve the imputation accuracy, TimeMultiformer incorporates three key mechanisms: (1) It takes the time lag information Δ as part of the input to capture the temporal dependencies in sequential data. (2) It adopts the joint data imputation learning strategy (manually masking some observed values) to

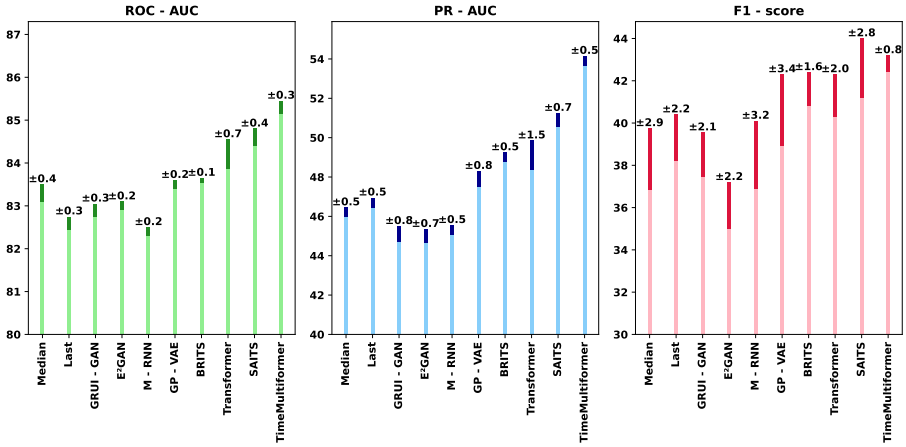

Fig. 2. Downstream Classification Results on PhysioNet-2012 Metrics are based on five independent runs, reported as means ± standard deviations. Light colors denote means, and dark regions represent standard deviations in the figure. Higher values indicate better performance.

enhance the model's robustness against incomplete data. (3) It uses the diagonally masked multi-head attention mechanism in imputation to effectively model the long-range dependencies within the sequence. In this subsection, we conduct ablation experiments to verify the effectiveness of each design in improving the missing value (MV) imputation accuracy.

Table 4 presents the ablation results on four real-world datasets. The baseline model excluding all three proposed designs (denoted as 'Base' in the table) exhibits the lowest imputation accuracy, confirming the necessity of these mechanisms. Starting from the base model, iterative integration of each design progressively enhances performance, with TimeMultiformer achieving optimal results once all components are combined. These findings validate the individual contributions of each design in improving missing value (MV) imputation accuracy.

Additionally, we implemented a single iTransformer, serving as the feature dependencies learning net component of TimeMultiformer. As shown in Table 5, TimeMultiformer exhibits further performance gains when integrated with the temporal capture network component.

6 Conclusion

This paper proposes TimeMultiformer, an attention-based model that enhances time series imputation accuracy through the collaborative design of feature and temporal dependencies learning net. Trained with a joint optimization strategy, the model outperforms existing methods in imputation tasks. When compared

Table 5. Ablation Experiment Results without Temporal Dependencies Learning Net

Model	PhysioNet-2012	Air-Quality
iTransformer	0.299 / 0.436 / 44.6%	0.134 / 0.339 / 19.1%
TimeMultiformer	**0.189** / **0.327** / **28.5%**	**0.118** / **0.303** / **16.7%**
Model	ETT(m1)	Electricity
iTransformer	0.144 / 0.212 / 13.8%	0.588 / 0.988 / 31.2%
TimeMultiformer	**0.084** / **0.130** / **8.0%**	**0.572** / **0.945** / **30.6%**

to state-of-the-art (SOTA) models, TimeMultiformer significantly reduces errors across four real-world datasets: mean absolute error (MAE) by 2%–16%, root mean square error (RMSE) by 10%–26%, and mean relative error (MRE) with an average reduction exceeding 10%, achieving the current optimal imputation accuracy for time series data. Downstream classification experiments further demonstrate that the complete data imputed by TimeMultiformer directly improves the performance of classification tasks. Despite the aforementioned advantages, TimeMultiformer still has room for further improvement. In the future, we will explore lightweight attention variants or dynamic pruning strategies to reduce redundant computations while maintaining imputation accuracy. Secondly, the current framework assumes that the missing patterns in time series are random, but actual data often contain non-random or structured missingness. In future work, we will also consider more missing patterns.

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